Working with Arduino

This chapter contains the following sections. Please read as needed:

- Arduino Getting Started
- Setting Up the Development Environment
- Example

Arduino Getting Started

New to Arduino ESP32 development and looking for a quick start? We have prepared a comprehensive **Getting Started Tutorial** for you.

- Section 0: Getting to Know ESP32
- Section 1: Installing and Configuring Arduino IDE
- Section 2: Arduino Basics
- Section 3: Digital Output/Input
- Section 4: Analog Input
- Section 5: Pulse Width Modulation (PWM)
- Section 6: Serial Communication (UART)
- Section 7: I2C Communication
- Section 8: SPI Communication
- Section 9: Wi-Fi Basics
- Section 10: Web Server
- Section 11: Bluetooth
- Section 12: LVGL GUI Development
- Section 13: Comprehensive Project

Note: This tutorial uses the ESP32-S3-Zero as a reference example, and all hardware code is based on its pinout. Before you start, we recommend checking the pinout of your development board to ensure the pin configuration is correct.

Setting Up the Development Environment

1. Installing and Configuring the Arduino IDE

Please refer to the tutorial **Installing and Configuring Arduino IDE** to download and install the Arduino IDE and add ESP32 support.

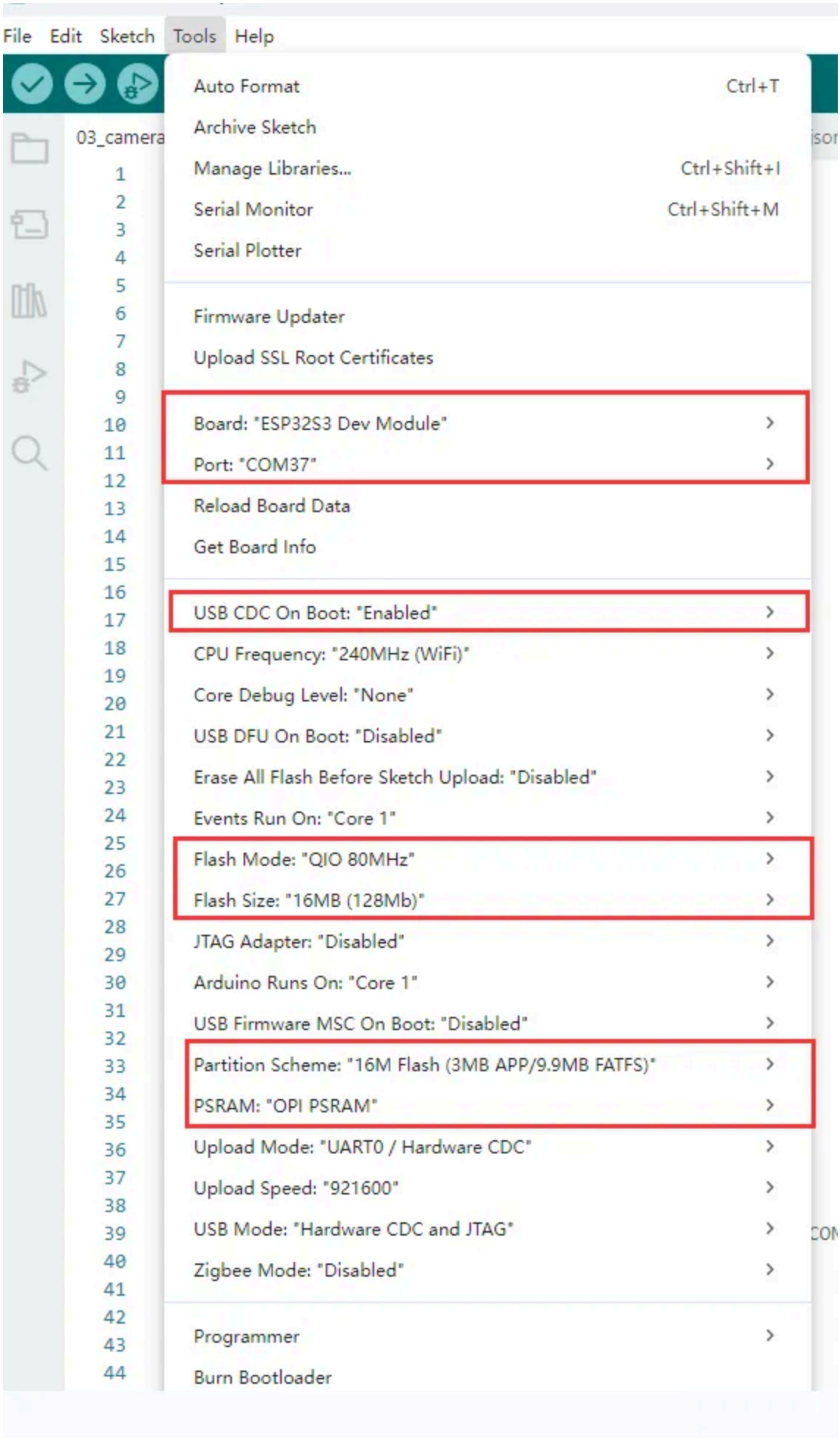
2. Installing Libraries

- When installing Arduino libraries, there are typically two methods: online installation and offline installation. If the library installation requires offline installation, you must use the provided library files.
- For most libraries, users can easily search for and install them via the Arduino IDE's online Library Manager. However, some open-source or custom libraries are not synchronized to the Arduino Library Manager and therefore cannot be found through online search. In this case, users can only install these libraries manually via offline methods.
- The example package for the ESP32-S3-Touch-LCD-3.5B development board can be downloaded from here. The `Arduino\libraries` directory within the package already contains all the library files required for this tutorial.

Library/File Name	Description	Version	Installation Method
LVGL	Graphics library	v8.4.0	"Online installation" (requires copying the demos folder to src)
GFX_Library_for_Arduino	LCD driver library	v1.5.5	"Online Installation"
XPowersLib	Power management driver library	v0.2.9	"Online Installation"

Library/File Name	Description	Version	Installation Method
SensorLib	Sensor library	v0.3.1	"Online Installation"
ESP32-audioI2S-master	Audio processing	v3.3.0	"Online Installation"
TCA9554	I/O expander library	v0.1.2	"Online Installation"
OneButton	Button library	v2.6.1	"Online Installation"
esp_lcd_touch_axs15231b	Touch driver	-	"Offline Installation"
es8311	es8311 driver library	-	"Offline Installation"

3. Arduino Project Parameter Settings


File Edit Sketch Tools Help
03_camera
Auto Format Ctrl+T
Archive Sketch
Manage Libraries... Ctrl+Shift+I
Serial Monitor Ctrl+Shift+M
Serial Plotter
Firmware Updater
Upload SSL Root Certificates
Board: "ESP32S3 Dev Module"
Port: "COM37"
Reload Board Data
Get Board Info
USB CDC On Boot: "Enabled"
CPU Frequency: "240MHz (WiFi)"
Core Debug Level: "None"
USB DFU On Boot: "Disabled"
Erase All Flash Before Sketch Upload: "Disabled"
Events Run On: "Core 1"
Flash Mode: "QIO 80MHz"
Flash Size: "16MB (128Mb)"
JTAG Adapter: "Disabled"
Arduino Runs On: "Core 1"
USB Firmware MSC On Boot: "Disabled"
Partition Scheme: "16M Flash (3MB APP/9.9MB FATFS)"
PSRAM: "OPI PSRAM"
Upload Mode: "UART0 / Hardware CDC"
Upload Speed: "921600"
USB Mode: "Hardware CDC and JTAG"
Zigbee Mode: "Disabled"
Programmer
Burn Bootloader

Example

The Arduino examples are located in the `Arduino/examples` directory of the example package.

Example	Basic Program Description	Dependency Library
01_audio_out	Read audio file from TF card and play it	ESP32-audioI2S-master
02_axp2101_example	Print data from the power management chip	XPowersLib
03_button_example	Single-click, double-click, long-press, etc., on the BOOT button	OneButton
04_es8311_example	Record audio for a period of time and play it back	es8311
05_pcf85063_example	Print RTC data	SensorLib
06_qmi8658_example	Print IMU data	SensorLib
07_sd_test	Test TF card read/write	-
08_gfx_helloworld	Display HelloWorld on the screen	GFX_Library_for_Arduino, TCA9554
09_lvgl_example_v8	LVGL v8 example program	lvgl (v8.4.0), GFX_Library_for_Arduino, TCA9554, esp_lcd_touch_axs15231b
10_lvgl_example_v9	LVGL v9 example program	lvgl (v9.2.2), GFX_Library_for_Arduino, TCA9554, esp_lcd_touch_axs15231b

Example	Basic Program Description	Dependency Library
11_camera_web_server	Display camera image on a web page	XPowersLib

01_audio_out

Example Description

- This example demonstrates the ESP32-S3-Touch-LCD-3.5B reading an audio file from the TF card and playing it through a speaker, supporting MP3, AAC, WAV, and other formats.

Hardware Connection

- Insert the TF card into the computer.
- Copy the music file `1.mp3` to the TF card.
- Insert the TF card into ESP32-S3-Touch-LCD-3.5B

Code Analysis

- Set the audio file to play

```
audio.connecttoFS(SD_MMC, "1.mp3");
```

Operation Result

- The device will play auido directly without showing content on the screen

02_axp2101_example

Example Description

- This example demonstrates using XPowers to drive the AXP2101 and printing data via the serial port.

Hardware Connection

- Connect the board to your computer via USB.

Code Analysis

- Initialize

```
bool result = power.begin(Wire, AXP2101_SLAVE_ADDRESS, i2c_sda, i2c_scl);

if (result == false) {
  Serial.println("power is not online...");
  while (1) delay(50);
}
```

Operation Result

- The screen shows no output
- Open the serial monitor



03_button_example

Example Description

- This example demonstrates using the OneButton library to drive the BOOT button and prints events such as single-click, double-click, and long-press via the serial port.

Hardware Connection

- Connect the board to your computer via USB.

Code Analysis

- Bind callback functions

```
button1.attachClick(click1);
button1.attachDoubleClick(doubleclick1);
button1.attachLongPressStart(longPressStart1);
button1.attachLongPressStop(longPressStop1);
button1.attachDuringLongPress(longPress1);
```

Operation Result

- The screen shows no output
- Open the serial monitor

```
BOOT Button click.
BOOT Button doubleclick.
BOOT Button longPress start
BOOT Button longPress...
BOOT Button longPress...
BOOT Button longPress...
BOOT Button longPress...
```

04_es8311_example

Example Description

- This example demonstrates using the ESP32-S3-Touch-LCD-3.5B to drive the ES8311 audio codec to implement audio recording and playback.

Hardware Connection

- Connect the board to your computer via USB.

Code Analysis

- Initialize

```
Wire.begin(I2C_SDA, I2C_SCL);
es8311_codec_init();
setupI2S();
```

- Record audio for 5 seconds and play back the recorded content:

```
wav_buffer = i2s.recordWAV(5, &wav_size);
delay(1000);
Serial.println("I2S playWAV");
i2s.playWAV(wav_buffer, wav_size);
```

Operation Result

- The screen shows no output
- Press the RST button on the ESP32-S3-Touch-LCD-3.5B to start recording; playback begins 2 seconds later.

05_pcf85063_example

Example Description

- This example demonstrates driving the PCF85063 on the ESP32-S3-Touch-LCD-3.5B to set the time/date and read the time.

Hardware Connection

- Connect the board to your computer via USB.

Code Analysis

- Initialize

```
if (!rtc.begin(Wire, PCF85063_SLAVE_ADDRESS, SENSOR_SDA, SENSOR_SCL)) {
    Serial.println("Failed to find PCF8563 - check your wiring!");
    while (1) {
        delay(1000);
    }
}
```

- Set the time and date

```
rtc.setDateTime(year, month, day, hour, minute, second);
```

- Get the time and date

```
RTC_DateTime datetime = rtc.getDateTime();
```

Operation Result

- The screen shows no output
- Open the serial monitor

```
[UNIX]: Year :2025 Month:01 Day :23 Hour:15 Minute:02 Sec :03
Thursday, January 23 2025 15:02:03
Jan 23 2025 15:02:03
Thursday, 23. January 2025 03:02PM
[RTC ]: Year :2025 Month:01 Day :23 Hour:15 Minute:02 Sec :04
[UNIX]: Year :2025 Month:01 Day :23 Hour:15 Minute:02 Sec :04
Thursday, January 23 2025 15:02:04
Jan 23 2025 15:02:04
Thursday, 23. January 2025 03:02PM
[RTC ]: Year :2025 Month:01 Day :23 Hour:15 Minute:02 Sec :05
[UNIX]: Year :2025 Month:01 Day :23 Hour:15 Minute:02 Sec :05
Thursday, January 23 2025 15:02:05
Jan 23 2025 15:02:05
Thursday, 23. January 2025 03:02PM
```

06_qmi8658_example

Example Description

- This example demonstrates driving the QMI8658 on the ESP32-S3-Touch-LCD-3.5B to read and print accelerometer, gyroscope, and IMU temperature data.

Hardware Connection

- Connect the board to your computer via USB.

Code Analysis

- Initialize

```
if (!qmi.begin(Wire, QMI8658_L_SLAVE_ADDRESS, SENSOR_SDA, SENSOR_SCL)) {
  Serial.println("Failed to find QMI8658 - check your wiring!");
  while (1) {
    delay(1000);
  }
}
```

- Get data

```
if (qmi.getDataReady()) {
  if (qmi.getAccelerometer(acc.x, acc.y, acc.z)) {
  }
  if (qmi.getGyroscope(gyr.x, gyr.y, gyr.z)) {
  }
}
```

Operation Result

- The screen shows no output
- Open the serial monitor

```
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 95-----GYRO. x:5. 79, GYRO. y:-5. 37, GYRO. z:-0. 00-----Temperature:31. 25 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 94-----GYRO. x:5. 48, GYRO. y:-5. 12, GYRO. z:0. 61-----Temperature:31. 22 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 94-----GYRO. x:5. 24, GYRO. y:-5. 02, GYRO. z:0. 15-----Temperature:31. 24 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 94-----GYRO. x:6. 05, GYRO. y:-5. 60, GYRO. z:-0. 01-----Temperature:31. 27 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 95-----GYRO. x:6. 23, GYRO. y:-5. 13, GYRO. z:0. 08-----Temperature:31. 24 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 94-----GYRO. x:5. 97, GYRO. y:-5. 45, GYRO. z:0. 04-----Temperature:31. 25 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 95-----GYRO. x:5. 58, GYRO. y:-5. 39, GYRO. z:0. 19-----Temperature:31. 23 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 94-----GYRO. x:5. 45, GYRO. y:-5. 25, GYRO. z:-0. 31-----Temperature:31. 25 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 18, ACCEL. z:-0. 95-----GYRO. x:5. 37, GYRO. y:-4. 48, GYRO. z:-0. 11-----Temperature:31. 23 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 95-----GYRO. x:5. 05, GYRO. y:-5. 42, GYRO. z:0. 10-----Temperature:31. 25 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 95-----GYRO. x:6. 03, GYRO. y:-5. 30, GYRO. z:0. 21-----Temperature:31. 26 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 94-----GYRO. x:5. 14, GYRO. y:-5. 62, GYRO. z:-0. 02-----Temperature:31. 21 degrees C
ACCEL. x:0. 01, ACCEL. y:0. 17, ACCEL. z:-0. 95-----GYRO. x:5. 80, GYRO. y:-5. 02, GYRO. z:0. 02-----Temperature:31. 25 degrees C
```

07_sd_test

Example Description

- This example demonstrates testing TF card read/write functionality on the ESP32-S3-Touch-LCD-3.5B.

Hardware Connection

- Connect the board to the computer.
- Insert the TF card into the card slot (TF card needs to be formatted as FAT32).

Code Analysis

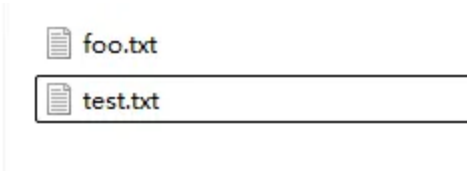
- Initialize

```
if(!SD_MMC.setPins(clk, cmd, d0)){
  Serial.println("Pin change failed!");
  return;
}

if (!SD_MMC.begin( "/sdcard", true)) {
  Serial.println("Card Mount Failed");
  return;
}
```

Operation Result

- Insert the TF card into the computer, and and two files, test.txt and foo.test, will appear. The content of the foo.txt is Hello World!, and the test.txt is empty.



08_gfx_helloworld

Example Description

- This example demonstrates using the GFX_Library_for_Arduino library to drive the screen and display "HelloWorld" on the screen.

Hardware Connection

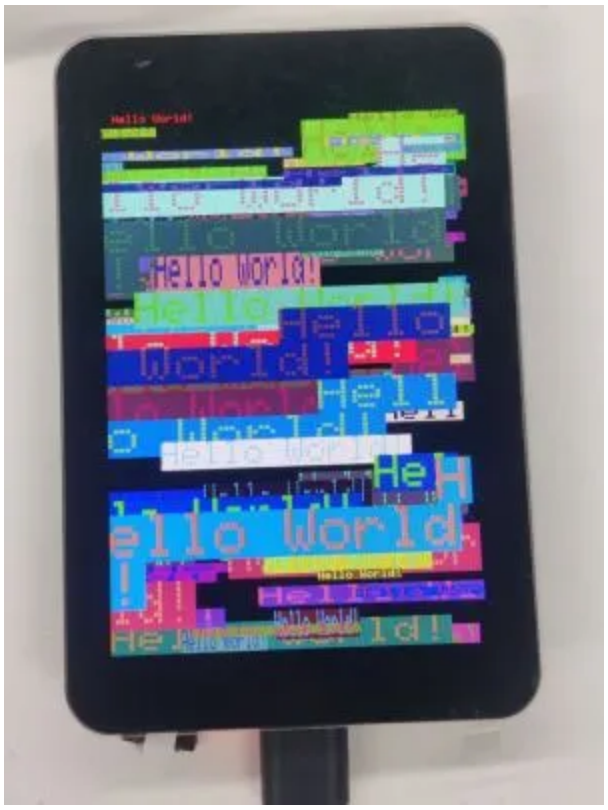
- Connect the board to the computer.

Code Analysis

- Configure the screen interface and resolution, etc.

```
  Arduino_DataBus *bus = new Arduino_ESP32QSPI(LCD_QSPI_CS, LCD_QSPI_CLK,
LCD_QSPI_D0, LCD_QSPI_D1, LCD_QSPI_D2, LCD_QSPI_D3);
  Arduino_GFX *g = new Arduino_AXS15231B(bus, -1 /* RST */, 0 /* rotation */,
false, 320, 480);
  Arduino_Canvas *gfx = new Arduino_Canvas(320, 480, g, 0, 0, ROTATION);
```

Operation Result



09_lvgl_example_v8

Example Description

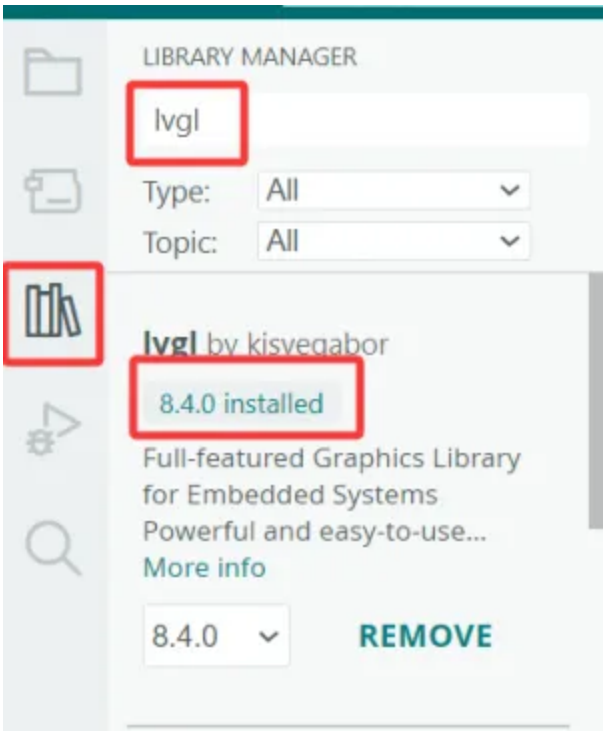
- This example demonstrates running the LVGL (v8.4.0) example program on the ESP32-S3-Touch-LCD-3.5B.

Hardware Connection

- Connect the board to the computer.

Preparation

- LVGL v8.4.0 must be installed. If another version is installed, please reinstall it.



Code Analysis

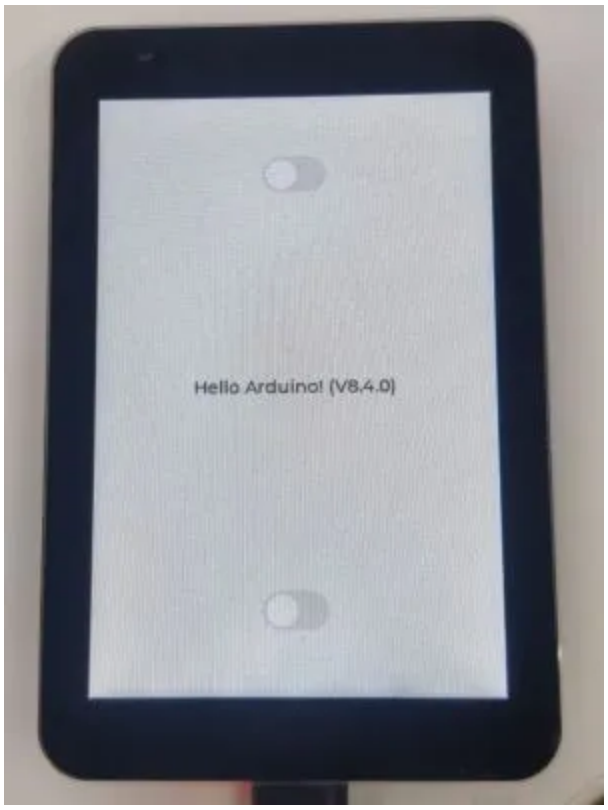
- UI initialization

```
  lv_obj_t *label = lv_label_create(lv_scr_act());
  lv_label_set_text(label, "Hello Arduino! (V" GFX_STR(LVGL_VERSION_MAJOR) "."
GFX_STR(LVGL_VERSION_MINOR) "." GFX_STR(LVGL_VERSION_PATCH) ")");
  lv_obj_align(label, LV_ALIGN_CENTER, 0, 0);

  lv_obj_t *sw = lv_switch_create(lv_scr_act());
  lv_obj_align(sw, LV_ALIGN_TOP_MID, 0, 50);

  sw = lv_switch_create(lv_scr_act());
  lv_obj_align(sw, LV_ALIGN_BOTTOM_MID, 0, -50);
```

Operation Result



10_lvgl_example_v9

Example Description

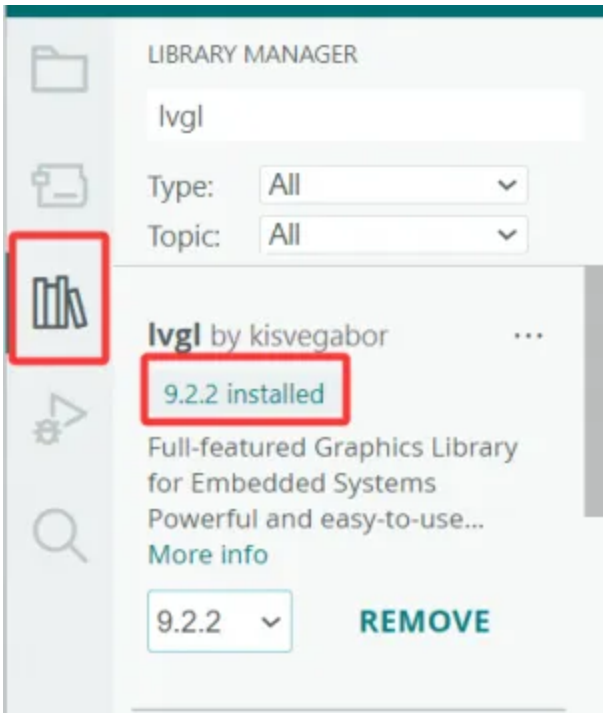
- This example demonstrates running the LVGL (v9.2.2) example program on the ESP32-S3-Touch-LCD-3.5B.

Hardware Connection

- Connect the board to the computer.

Preparation

- LVGL v9.2.2 must be installed. If another version is installed, please reinstall it.



Code Analysis

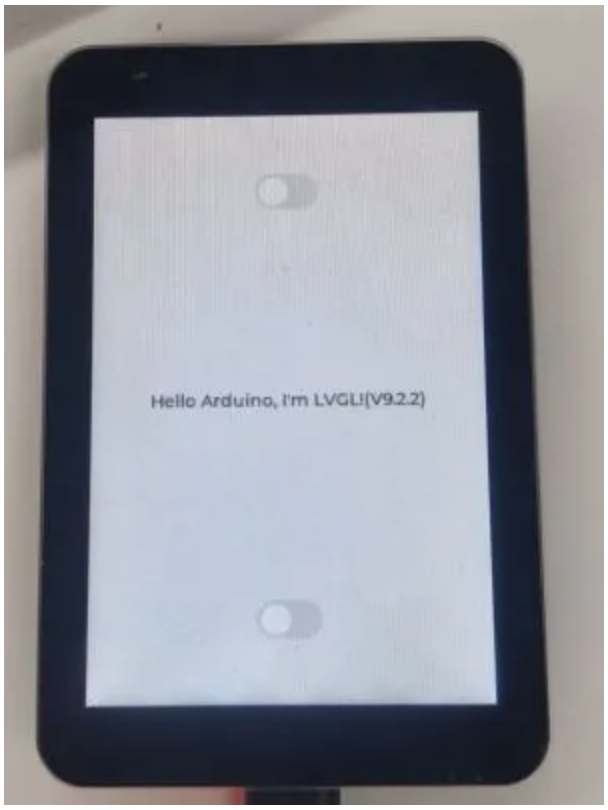
- UI initialization

```
  lv_obj_t *label = lv_label_create(lv_scr_act());
  lv_label_set_text(label, "Hello Arduino, I'm LVGL!(V"
GFX_STR(LVGL_VERSION_MAJOR) "." GFX_STR(LVGL_VERSION_MINOR) "."
GFX_STR(LVGL_VERSION_PATCH) ")");
  lv_obj_align(label, LV_ALIGN_CENTER, 0, 0);

  lv_obj_t *sw = lv_switch_create(lv_scr_act());
  lv_obj_align(sw, LV_ALIGN_TOP_MID, 0, 50);

  sw = lv_switch_create(lv_scr_act());
  lv_obj_align(sw, LV_ALIGN_BOTTOM_MID, 0, -50);
```

Operation Result



11_camera_web_server

Example Description

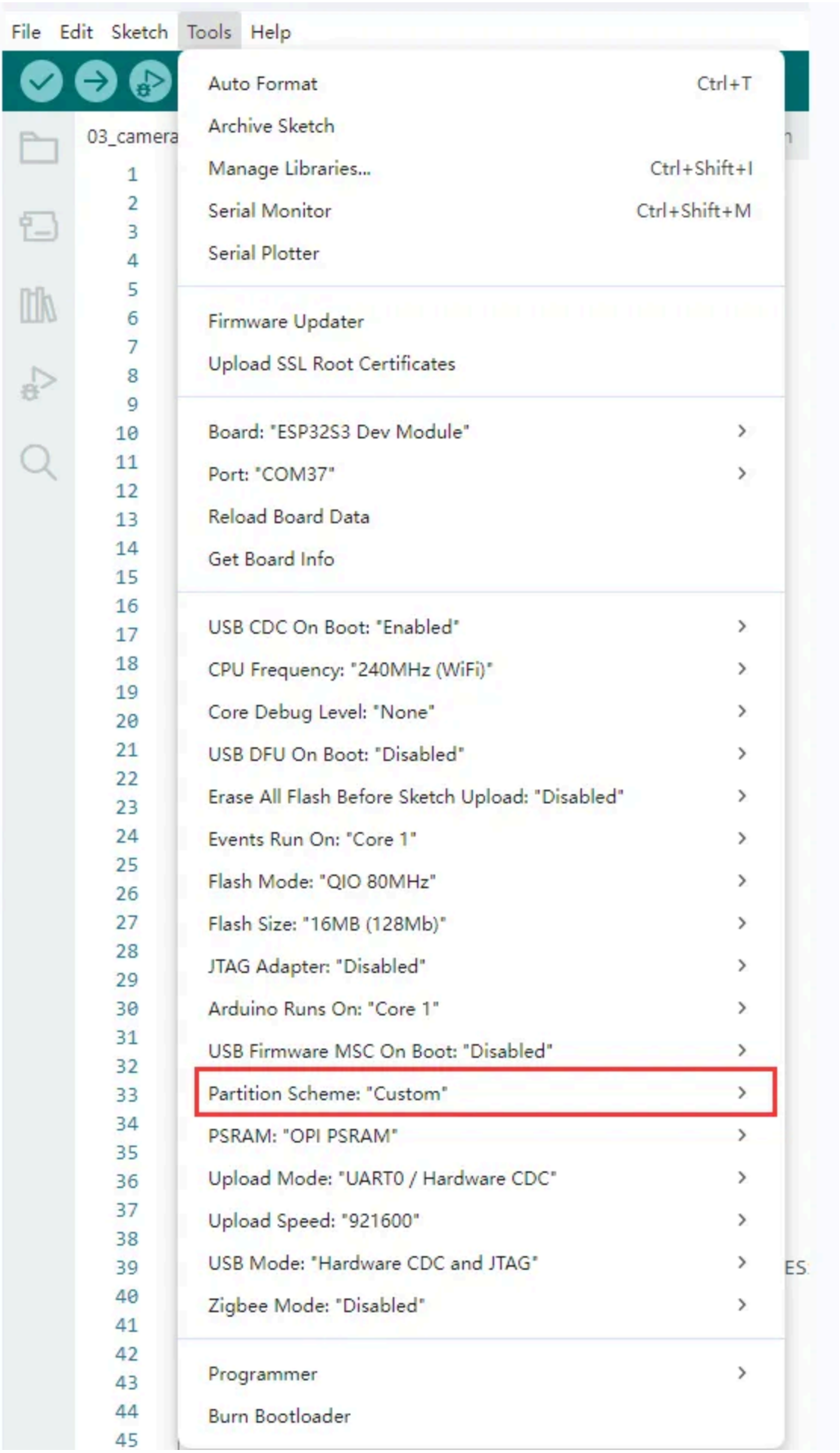
- This example demonstrates how to drive the camera. After connecting to Wi-Fi, the program creates a web server. Users simply enter the device's IP address in a browser to access it. The web page can display the camera image and supports settings such as resolution and mirroring.

Hardware Connection

- Connect the board to the computer.
- Insert the OV5640 camera into the 24Pin header on the board (not required for the version with a case).

Preparation

- Set Parttion to Custom.


File Edit Sketch Tools Help
03_camera
Auto Format Ctrl+T
Archive Sketch
Manage Libraries... Ctrl+Shift+I
Serial Monitor Ctrl+Shift+M
Serial Plotter
Firmware Updater
Upload SSL Root Certificates
Board: "ESP32S3 Dev Module"
Port: "COM37"
Reload Board Data
Get Board Info
USB CDC On Boot: "Enabled"
CPU Frequency: "240MHz (WiFi)"
Core Debug Level: "None"
USB DFU On Boot: "Disabled"
Erase All Flash Before Sketch Upload: "Disabled"
Events Run On: "Core 1"
Flash Mode: "QIO 80MHz"
Flash Size: "16MB (128Mb)"
JTAG Adapter: "Disabled"
Arduino Runs On: "Core 1"
USB Firmware MSC On Boot: "Disabled"
Partition Scheme: "Custom"
PSRAM: "OPI PSRAM"
Upload Mode: "UART0 / Hardware CDC"
Upload Speed: "921600"
USB Mode: "Hardware CDC and JTAG"
Zigbee Mode: "Disabled"
Programmer
Burn Bootloader

Code Analysis

- Define camera-related pins

```
#define PWDN_GPIO_NUM -1
#define RESET_GPIO_NUM -1
#define XCLK_GPIO_NUM 38
#define SIOD_GPIO_NUM 8
#define SIOC_GPIO_NUM 7

#define Y9_GPIO_NUM 21
#define Y8_GPIO_NUM 39
#define Y7_GPIO_NUM 40
#define Y6_GPIO_NUM 42
#define Y5_GPIO_NUM 46
#define Y4_GPIO_NUM 48
#define Y3_GPIO_NUM 47
#define Y2_GPIO_NUM 45
#define VSYNC_GPIO_NUM 17
#define HREF_GPIO_NUM 18
#define PCLK_GPIO_NUM 41
```

- Wi-Fi SSID and password to connect to

```
const char *ssid = "waveshare";
const char *password = "12345678";
```

- Start the server to begin providing an HTTP video stream

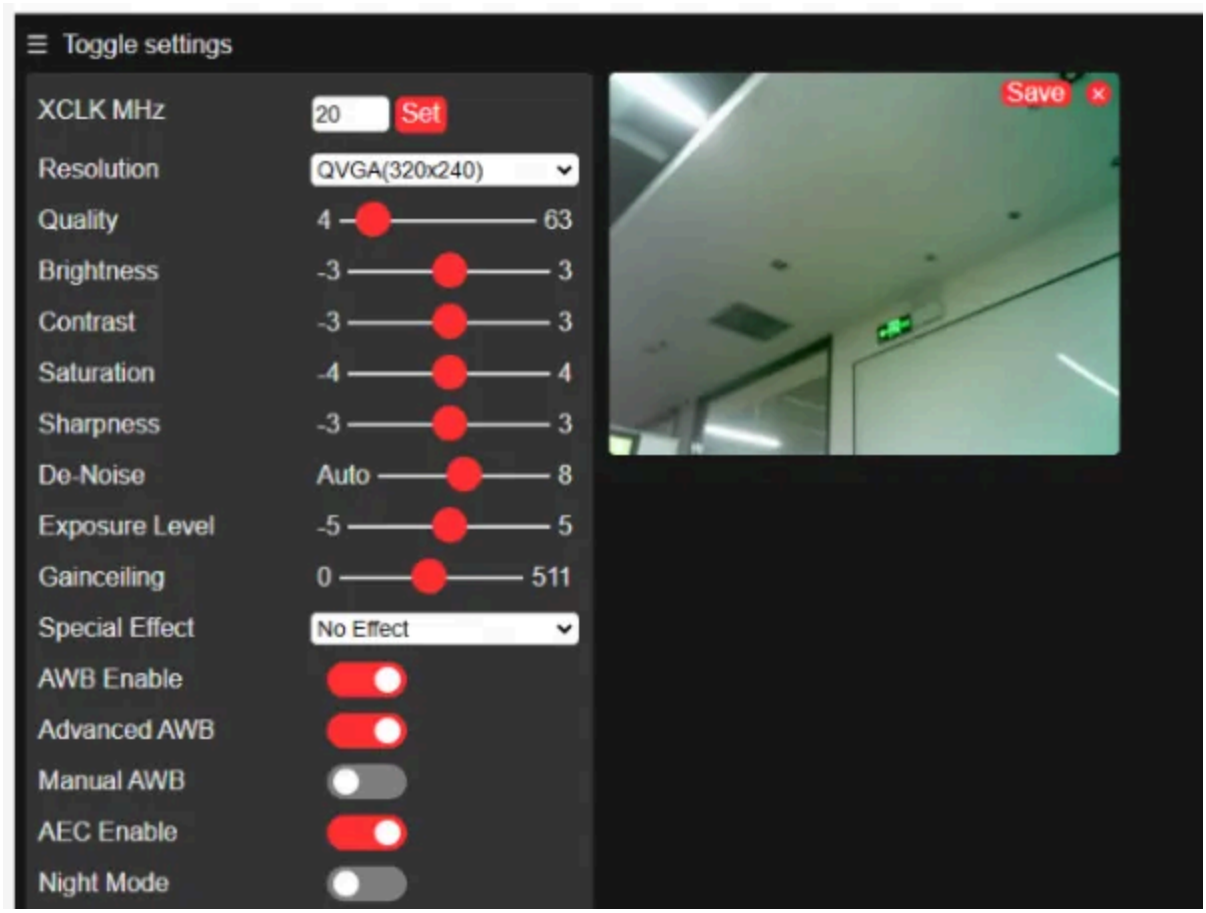
```
startCameraServer();
Serial.print("Camera Ready! Use 'http://");
Serial.print(WiFi.localIP());
Serial.println("' to connect");
```

Operation Result

- Open the serial terminal; after connecting to Wi-Fi, the IP address is shown.



- Enter the IP address printed on the serial port in a browser (on the device that provides the hotspot).
- Click "Start Stream" to see the camera image.



Give Feedback